

ANANTHA KRISHNAN O.S.

Chennai, Tamil Nadu | +91-7448581970 | osananthakrishnan4@gmail.com | linkedin.com/in/anantha-krishnan-os | github.com/AnanthaKrishnan45

PROFESSIONAL SUMMARY

B.Tech Computer Science (Artificial Intelligence & Machine Learning) graduate (2026) with hands-on experience in Python-based application development, AI/ML engineering, web applications, database management, API integration, and software evaluation. Strong foundation in software testing, debugging, problem-solving, data analysis, and application validation through academic and internship projects. Experienced with Python, SQL, MySQL, HTML, CSS, Git, GitHub, NLP, Generative AI, Retrieval-Augmented Generation (RAG), and cloud-based AI APIs. Familiar with functional, integration, regression testing, defect identification, test-result analysis, and root-cause analysis. Seeking an SDET Intern opportunity to develop automated testing solutions for web applications, AI/chatbot applications, and REST APIs.

TECHNICAL SKILLS

Programming Languages: Python, SQL, HTML, CSS

Software Testing: Software Testing Principles, Functional Testing, Integration Testing, Regression Testing, Application Validation, Debugging, Defect Identification, Test Result Analysis, Root-Cause Analysis

Web Technologies: HTML5, CSS3, Responsive Web Development, Web Applications

API & Cloud: API Integration, Microsoft Azure Cognitive Services, Microsoft Speech SDK, Cloud-Based Applications

Databases: MySQL, SQL, Database Querying, Data Analysis

AI & Generative AI: Generative AI, Retrieval-Augmented Generation (RAG), NLP, Machine Learning, Deep Learning, AI Application Development

Data & Analysis: Pandas, NumPy, Data Preprocessing, Feature Engineering, Exploratory Data Analysis, Data Visualization

Version Control & Tools: Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Kaggle

Development Concepts: Software Development Lifecycle, Application Testing, Model Validation, Performance Evaluation

Collaboration: Agile/Scrum Fundamentals, Team Collaboration, Communication, Adaptability, Quick Learning

EDUCATION

B.Tech – Computer Science (Artificial Intelligence & Machine Learning) | SRM Institute of Science and Technology, Ramapuram, Chennai | 2026 | **CGPA: 7.62/10**

Higher Secondary – CBSE | Narayana e-Techno School, Velappanchavadi, Chennai | 2022 | 72%

Secondary School – CBSE | Narayana e-Techno School, Pallavaram, Chennai | 2020 | 60%

EXPERIENCE

AI Intern | Internzvalley | June 2024

- Developed and evaluated Python-based machine learning solutions as part of end-to-end application workflows.
- Performed data preprocessing, feature engineering, exploratory data analysis, model training, validation, and performance tuning.
- Validated machine learning models using evaluation metrics and analyzed results to identify performance issues and improvement opportunities.
- Applied systematic debugging and problem-solving techniques during model development, testing, and validation.
- Worked with structured datasets and SQL-based data workflows for data analysis and application development.
- Analyzed application and model outputs to identify inconsistencies and improve solution quality.
- Gained practical experience with software development workflows, technical documentation, and collaborative project execution.

PROJECTS

RAG-Based AI Study Partner

Technologies: Python, LangChain, Retrieval-Augmented Generation (RAG), NLP, Generative AI, Vector Database

- Built an AI-powered study assistant capable of answering user queries from uploaded documents through a conversational application workflow.
- Developed a document ingestion and chunking pipeline to process user-provided knowledge sources.
- Implemented vector-based semantic retrieval to identify relevant information from custom knowledge bases.
- Integrated Generative AI with retrieved context to generate grounded responses and reduce unsupported outputs.
- Tested application responses against document context to evaluate relevance, factual consistency, and retrieval quality.
- Analyzed incorrect or unsupported responses and refined the retrieval and generation workflow to improve application reliability.
- Applied Gen AI concepts including semantic retrieval, contextual grounding, prompt-driven generation, and hallucination reduction.

Text-to-Speech Web Application

Technologies: Python, HTML5, CSS3, NLP, Microsoft Azure Cognitive Services, Microsoft Speech SDK

- Designed and developed a cloud-integrated web application combining a responsive HTML/CSS interface with NLP-based language processing.
- Integrated Microsoft Azure Cognitive Services and Speech SDK APIs to provide real-time text-to-speech functionality.
- Implemented context-aware pronunciation, emotion styling, and multilingual voice generation capabilities.
- Tested application functionality across different input scenarios to validate speech generation, language handling, and application behavior.
- Performed functional validation and troubleshooting of API-integrated application workflows.
- Analyzed application outputs and performance to identify issues and improve reliability.
- Implemented responsive UI behavior for web and mobile platforms and achieved 98% reported speech synthesis accuracy.

Uber Stock Price Prediction

Technologies: Python, Scikit-Learn, Pandas, Matplotlib, SQL, Kaggle

- Developed a machine learning regression application using historical financial datasets.
- Collected, cleaned, and validated datasets before model development.
- Performed feature engineering and exploratory data analysis to identify data quality and input-related issues.
- Used SQL and Python-based data processing techniques to query, analyze, and prepare structured data.
- Trained and evaluated a regression model, achieving 93% reported predictive accuracy.
- Analyzed model results and iteratively evaluated performance to improve prediction quality.

CERTIFICATIONS

-
- Data Science Master Class — Scaler | January 2025
 - Deep Learning Course — Scaler | October 2024
 - Artificial Intelligence — Internzvalley | June 2024

ACHIEVEMENTS

- Intel Hackathon Participant — KPR Institute of Engineering and Technology | October 2024

VOLUNTEER EXPERIENCE

Volunteer | Nizhal Foundation, Chennai | June 2024

- Participated in team-based community initiatives requiring coordination, communication, and collaborative execution.
- Contributed to community engagement and environmental conservation activities.